User's Documentation for Vulkan OBJ Previewer

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1 Introduction

Vulkan OBJ Previewer is a high-performance, cross-platform 3D model viewer designed for real-time inspection of geometry and materials. The application utilizes the Vulkan API to ensure low-overhead rendering and features a custom built parser for `.obj` and `.mtl` files, allowing for efficient loading of 3D assets without external heavy-duty libraries.

2 Installation and Setup

To run or build the application, ensure your system meets the following requirements:

2.1 Prerequisites

- **Vulkan SDK:** Must be installed and the `VULKAN_SDK` environment variable must be set.
- **Graphics Drivers:** Up-to-date drivers with Vulkan support.
- **C++ Compiler:** A compiler supporting C++20 (e.g., MSVC for Windows, Apple Clang for macOS).
- **CMake:** Version 3.15 or newer is required to configure the build.
- **OS Specifics:** Windows users should have `vcpkg` installed; macOS users require `Homebrew`.

2.1.1 Prerequisites installation

- **Vulkan SDK:** Vulkan SDK is installed from the website `vulkan.lunarg.com` depending on your system.
- **vcpkg:** For Windows machines the dependency script uses Microsoft's `vcpkg` manager, a guide on installing it is shown when running the dependency script; in order to run the dependency script without any hiccups the `vcpkg` path has to be added to the system environment variables, again deeply explained when running the dependency script even though this package is missing.

2.2 Launching the Application

Depending on your operating system, the executable is located in the following directories after building:

- **Windows:** `build/Debug/VulkanApp.exe` or `build/Release/VulkanApp.exe`.
- **macOS:** `build/VulkanApp.app/Contents/MacOS/VulkanApp`.

3 Controls and Navigation

The application offers an intuitive interface using both mouse and keyboard/menu interactions.

3.1 Interaction

- **Rotate Camera:** Click and drag the Left Mouse Button.
- **Zoom In / Out:** Use the Mouse Wheel.

3.2 Native Menu Actions

Users can access additional features via the native OS menu bar:

- **File → Open .obj File:** Select and load a new model from disk.
- **View → Toggle Wireframe:** Switch between solid and wireframe rendering modes.
- **View → Switch Light Source:** Re-orient directional lighting to the current camera view.
- **View → Zoom Sensitivity:** Adjust the scrolling speed.

4 Working with 3D Models

The previewer is optimized for the Wavefront `.obj` format.

- **Geometry:** Supports vertex positions, normals, and texture coordinates.
- **Materials:** Associated `.mtl` files are parsed to apply material properties and textures.
- **Lighting:** The application uses dynamic shadow mapping for realistic depth.

5 Troubleshooting

If the application fails to start or render correctly, check the following:

- **Shader Compilation:** Ensure that `compile-shaders.sh` has been executed to generate SPIR-V bytecode.
- **Validation Layers:** If running in Debug mode, ensure Vulkan validation layers are properly installed. Debug/Release mode are enabled with standard CMake macros.
- **GPU Support:** Verify that your hardware supports the Vulkan API.

6 User Interface Gallery

Below are examples of the Vulkan OBJ Previewer running on both supported operating systems.

Figure 1: Application interface running on macOS.

Figure 2: Application interface running on Windows.

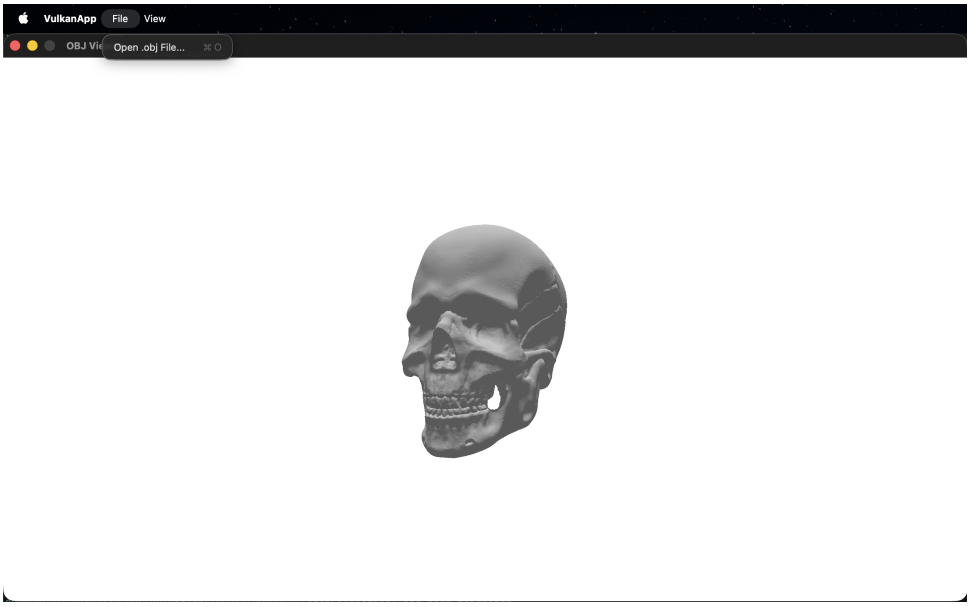

Figure 3: File menu on macOS

Figure 4: File menu on Windows

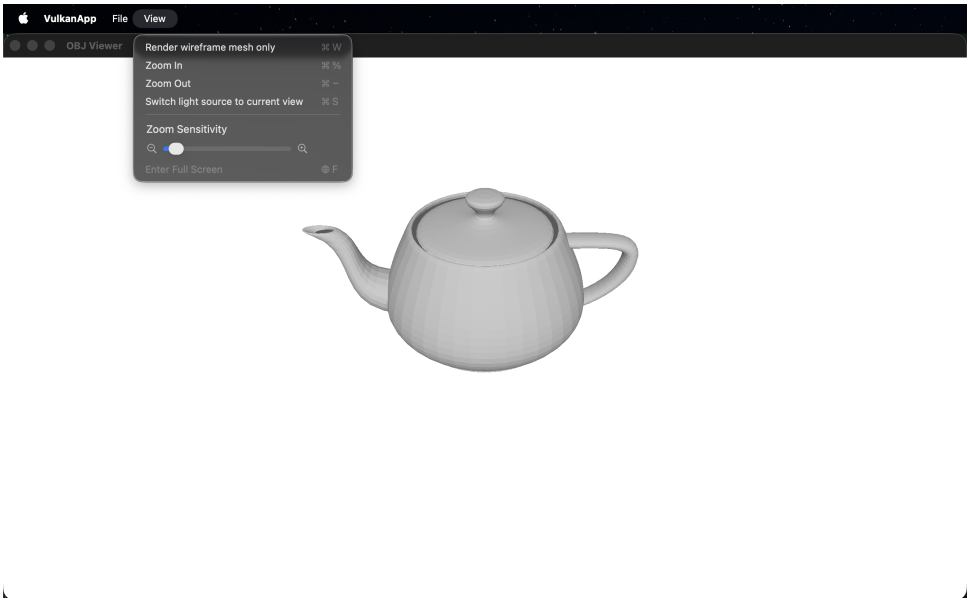

Figure 5: View menu on macOS

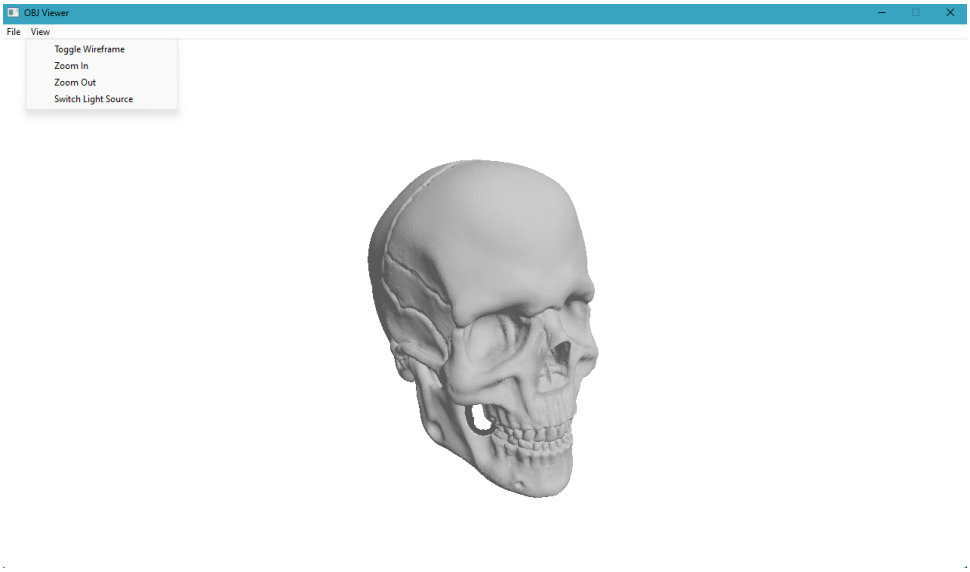

Figure 6: View menu on Windows